With transformation:

a) $X=\log(Y)$, $Y>0$.

$f(x) \propto \left(1+\left(\frac{x-\lambda}{\alpha}\right)^2\right)^{-m} e^{-\nu \tan^{-1}\left(\frac{x-\lambda}{\alpha}\right)}$.

$y=\tan^{-1}\left(\frac{x-\lambda}{\alpha}\right)$. $\quad y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.

$\tan y = \frac{x-\lambda}{\alpha}$ $\quad$ ↳ finite domain

$= (1+(\tan y)^2)^{-m} e^{-\nu y}$

$= (\sec^2(y))^{-m} e^{-\nu y}$

$= (\cos^2(y))^{m} e^{-\nu y}$ $\quad$ $\left.\begin{array}{l} m>\frac{3}{2} \\ \nu \in \mathbb{R} \end{array}\right\}$ shape $\quad$ $\left.\begin{array}{l} \lambda \\ \alpha>0 \end{array}\right\}$ scale.

$= (\cos(y))^{2m} e^{-\nu y}$.

① Uniform.

$f(y) = K(\cos(y))^{2m} e^{-\nu y}$. $\quad Y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.

Try

$g(y) = \frac{1}{b-a} = \frac{1}{\pi}$. The Y can be simulated by $\pi \times U - \frac{\pi}{2}$, $U \sim \text{Unif}[0,1]$.

$\frac{f(y)}{cg(y)} \leq 1$.

$\frac{f(y)}{g(y)} \leq c$.

$\pi K(\cos(y))^{2m} e^{-\nu y} \leq c$.

$\frac{d}{dy}\left[\pi K(\cos(y))^{2m} e^{-\nu y}\right] = 0$.

$\frac{d}{dy}\left[\cos(y)^{2m} e^{-\nu y}\right] = 0$.

$-2m e^{-\nu y}(\cos y)^{2m-1}\sin(y) - \nu e^{-\nu y}(\cos y)^{2m} = 0$. $\quad \frac{d}{dx}((x+1)^{-1}) \quad x \geq 0$.

$\quad = -\frac{1}{(x+1)^2} \neq 0$.

$e^{-\nu y}(\cos y)^{2m} = 0$. $\quad -2m\tan(y) - \nu = 0$.

$y = \pm\frac{\pi}{2}$. $\quad \tan(y) = -\frac{\nu}{2m}$.

$y = \tan^{-1}\left(-\frac{\nu}{2m}\right)$.

Hence, take $c = \pi K\left(\cos\left(\tan^{-1}\left(-\frac{\nu}{2m}\right)\right)\right)^{2m} e^{-\nu \tan^{-1}\left(-\frac{\nu}{2m}\right)}$.

Accept if

$$u < \frac{\cancel{K}(\cos(Y))^{2m}e^{-\nu Y}}{\cancel{\pi K}(\cos(\tan^{-1}(-\frac{\nu}{2m})))^{2m} e^{-\nu\tan^{-1}(-\frac{\nu}{2m})} \cancel{\times\frac{1}{\pi}}}$$

$$= \frac{(\cos(Y))^{2m}e^{-\nu Y}}{(\cos(\tan^{-1}(-\frac{\nu}{2m})))^{2m} e^{-\nu\tan^{-1}(-\frac{\nu}{2m})}}$$

$Y \sim \text{Unif}[-\frac{\pi}{2}, \frac{\pi}{2}]$.

$(\cos(y))^2$ Can't be used, as we are unable to use inverse transform algorithm.

$f(y) = K(\cos(y))^{2m}e^{-\nu y}$, $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.

Try

$g(y) = A(\cos y)^2$

Find A:

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} A(\cos y)^2\, dy = 1.$$

$$\frac{A}{2}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1+\cos 2y)\,dy = 1.$$

$$\frac{A}{2}\left[y + \frac{1}{2}\sin(2y)\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = 1.$$

$$\frac{A}{2}\left(\frac{\pi}{2} + \frac{1}{2}\sin\pi + \frac{\pi}{2} - \frac{1}{2}\sin(-\pi)\right) = 1.$$

$$\frac{A\pi}{2} = 1$$

$$A = \frac{2}{\pi}.$$

Hence, $g(y) = \frac{2}{\pi}\cos^2 y$.

Find c:

$$\frac{f(y)}{g(y)} \leq c.$$

$$\frac{K(\cos(y))^{2m}e^{-\nu y}}{\frac{2}{\pi}\cos^2 y} \leq c.$$

$$\frac{\pi K}{2}(\cos y)^{2m-2}e^{-\nu y} \leq c.$$

Generate Y using inversed transform algorithm:

$$G(y) = \int_{-\frac{\pi}{2}}^{y} A(\cos t)^2\,dt$$

$$= \frac{A}{2}\left[t + \frac{1}{2}\sin(2t)\right]_{-\frac{\pi}{2}}^{y}$$

$$= \frac{A}{2}\left(y + \frac{1}{2}\sin(2y) + \frac{\pi}{2} - \frac{1}{2}\sin(-\pi)\right)$$

$$= \frac{A}{2}\left(y + \frac{1}{2}\sin(2y) + \frac{\pi}{2}\right).$$

$$= \frac{1}{\pi}\left(y + \frac{1}{2}\sin(2y) + \frac{\pi}{2}\right).$$

$$\frac{d}{dy}\left[\frac{\pi K}{2}(\cos y)^{2m-2}e^{-\nu y}\right] = 0$$

$$(2m-2)(-\sin y)(\cos y)^{2m-3}e^{-\nu y} + (\cos y)^{2m-2}(-\nu)e^{-\nu y} = 0.$$

$(\cos y)^{2m-2}e^{-\nu y} = 0$. $(2m-2)(-\sin y)(\cos y)^{-1} - \nu = 0.$

$y = \pm\frac{\pi}{2}.$

$(2m-2)\tan y = -\nu$

$y = \tan^{-1}(-\frac{\nu}{2m-2}).$

Hence, pick

$$c = \frac{\pi K}{2}\left[\cos(\tan^{-1}(-\frac{\nu}{2m-2}))\right]^{2m-2} e^{-\nu\tan^{-1}(-\frac{\nu}{2m-2})}$$

As a result, accept when

$$u < \frac{\cancel{K}(\cos(y))^{2m} e^{-\nu y}}{\cancel{\frac{\pi K}{2}}\left[\cos\left(\tan^{-1}\left(-\frac{\nu}{2m-2}\right)\right)\right]^{2m-2} e^{-\nu \tan^{-1}\left(-\frac{\nu}{2m-2}\right)} \cdot \cancel{\frac{2}{\pi}}\cos^2 y}$$

$$= \frac{(\cos y)^{2m-2} e^{-\nu y}}{\left[\cos\left(\tan^{-1}\left(-\frac{\nu}{2m-2}\right)\right)\right]^{2m-2} e^{-\nu \tan^{-1}\left(-\frac{\nu}{2m-2}\right)}}.$$

② $e^{-\nu y}$. Can't be used, will generate error.

$f(y) = K(\cos(y))^{2m} e^{-\nu y}$. $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.

Try

$g(y) = Ae^{-\nu y}$.

$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} Ae^{-\nu y}\, dy = 1$.

$-\frac{A}{\nu}\left[e^{-\nu y}\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = 1$.

$-\frac{A}{\nu}\left[e^{-\frac{\pi}{2}\nu} - e^{+\frac{\pi}{2}\nu}\right] = 1$.

$$A = \frac{\nu}{e^{\frac{\pi}{2}\nu} - e^{-\frac{\pi}{2}\nu}}$$

Find c:

$\frac{f(y)}{g(y)} \le c$.

$K\frac{e^{\frac{\pi}{2}\nu} - e^{-\frac{\pi}{2}\nu}}{\nu}(\cos(y))^{2m} \le c$.

$\frac{d}{dy}\left[K\frac{e^{\frac{\pi}{2}\nu} - e^{-\frac{\pi}{2}\nu}}{\nu}(\cos(y))^{2m}\right] = 0$.

$2m\sin y(\cos(y))^{2m-1} = 0$.

$\sin y = 0$. $\qquad \cos(y) = 0$.

$y = 0$ ✓ $\qquad y = \pm\frac{\pi}{2}$.

Hence, pick

$$c = K\frac{e^{\frac{\pi}{2}\nu} - e^{-\frac{\pi}{2}\nu}}{\nu}$$

Accept if

$$u < \frac{\cancel{K}(\cos(y))^{2m}\cancel{e^{-\nu y}}}{\cancel{K\frac{e^{\frac{\pi}{2}\nu} - e^{-\frac{\pi}{2}\nu}}{\nu}}\cancel{\frac{\nu}{e^{\frac{\pi}{2}\nu} - e^{-\frac{\pi}{2}\nu}}}\cancel{e^{-\nu y}}}.$$

$= (\cos(y))^{2m}$.

Inverse transform algorithm on $g(y)$.

$G(y) = \int_{-\frac{\pi}{2}}^{y} Ae^{-\nu z}\, dz$

$= -\frac{A}{\nu}\left[e^{-\nu z}\right]_{-\frac{\pi}{2}}^{y}$

$= -\frac{A}{\nu}\left(e^{-\nu y} - e^{+\frac{\pi}{2}\nu}\right)$.

$= \frac{1}{e^{-\frac{\pi}{2}\nu} - e^{\frac{\pi}{2}\nu}}\left(e^{-\nu y} - e^{\frac{\pi}{2}\nu}\right)$.

$u = \frac{1}{e^{-\frac{\pi}{2}\nu} - e^{\frac{\pi}{2}\nu}}\left(e^{-\nu y} - e^{\frac{\pi}{2}\nu}\right)$.

$e^{-\nu y} = u\left(e^{-\frac{\pi}{2}\nu} - e^{\frac{\pi}{2}\nu}\right) + e^{\frac{\pi}{2}\nu}$.

$y = -\frac{1}{\nu}\log\left[u\left(e^{-\frac{\pi}{2}\nu} - e^{\frac{\pi}{2}\nu}\right) + e^{\frac{\pi}{2}\nu}\right]$

$F^{-1}(u) = -\frac{1}{\nu}\log\left[u\left(e^{-\frac{\pi}{2}\nu} - e^{\frac{\pi}{2}\nu}\right) + e^{\frac{\pi}{2}\nu}\right]$

(3) $\cos(y)$

$f(y) = K(\cos(y))^{2m} e^{-\nu y}$, $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.

Try

$g(y) = A\cos y$

$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} A\cos y = 1.$

$A[\sin y]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = 1.$

$A(1-(-1)) = 1.$

$A = \frac{1}{2}.$

Hence, $g(y) = \frac{1}{2}\cos y$.

Find c:

$\frac{f(y)}{g(y)} \leq c.$

$\frac{K(\cos(y))^{2m} e^{-\nu y}}{\frac{1}{2}\cos y} \leq c.$

Inverse transform algorithm for $g(y)$:

$\int_{-\frac{\pi}{2}}^{y} A\cos z\,dz$

$= A[\sin z]_{-\frac{\pi}{2}}^{y}$

$= A(\sin y + 1)$

$= \frac{1}{2}(\sin y + 1).$

$U = \frac{1}{2}(\sin y + 1)$

$\sin y = 2U - 1$

$y = \sin^{-1}(2U-1)$

$F^{-1}(U) = \sin^{-1}(2U-1).$

$\frac{d}{dy}[2K(\cos y)^{2m-1} e^{-\nu y}] = 0.$

$\frac{d}{dy}((\cos y)^{2m-1} e^{-\nu y}) = 0.$

$(2m-1)(-\sin y)(\cos y)^{2m-2} e^{-\nu y} + (\cos y)^{2m-1}(-\nu)e^{-\nu y} = 0$

$(2m-1)\tan y + \nu = 0.$ $\quad (\cos y)^{2m-1} e^{-\nu y} = 0.$

$\tan y = -\frac{\nu}{2m-1}$ $\quad (\cos y)^{2m-1} = 0.$

$y = \tan^{-1}(-\frac{\nu}{2m-1}).$ $\quad y = \pm\frac{\pi}{2}.$

Thus, take

$c = 2K[\cos(\tan^{-1}(-\frac{\nu}{2m-1}))]^{2m-1} e^{-\nu\tan^{-1}(-\frac{\nu}{2m-1})}$,

Accept if

$$U < \frac{\cancel{K}(\cos(y))^{2m} e^{-\nu y}}{2\cancel{K}[\cos(\tan^{-1}(-\frac{\nu}{2m-1}))]^{2m-1} e^{-\nu\tan^{-1}(-\frac{\nu}{2m-1})} \cdot \cancel{\frac{1}{2}}\cos y}$$

$$= \frac{(\cos y)^{2m-1} e^{-\nu y}}{[\cos(\tan^{-1}(-\frac{\nu}{2m-1}))]^{2m-1} e^{-\nu\tan^{-1}(-\frac{\nu}{2m-1})}},$$

Without transformation.

a) $X=\log(Y),\ Y>0.$

$$f(x)\propto\left(1+\left(\frac{x-\lambda}{\alpha}\right)^2\right)^{-m} e^{-\nu\tan^{-1}\left(\frac{x-\lambda}{\alpha}\right)} \qquad x\in\mathbb{R}$$

④ $\left(1+\left(\frac{y-\lambda}{\alpha}\right)^2\right)^{-1}$. This is a Cauchy distribution.

$$f(y)=K\left(1+\left(\frac{y-\lambda}{\alpha}\right)^2\right)^{-m} e^{-\nu\tan^{-1}\left(\frac{x-\lambda}{\alpha}\right)}$$

$$g(y)=A\left(1+\left(\frac{y-\lambda}{\alpha}\right)^2\right)^{-1}.$$

$$\int_{-\infty}^{\infty} A\left(1+\left(\frac{y-\lambda}{\alpha}\right)^2\right)^{-1}dy=1$$

$$u=\frac{y-\lambda}{\alpha}.$$

$$\frac{du}{dx}=\frac{1}{\alpha}.\quad dx=\alpha du.$$

$$\int_{-\infty}^{\infty} A(1+u^2)^{-1}\alpha du. =1$$

$$A\alpha\int_{-\infty}^{\infty}\frac{1}{1+u^2}du=1.$$

$$A\alpha\left[\tan^{-1}(u)\right]_{-\infty}^{\infty}=1$$

$$A\alpha\pi=1$$

$$A=\frac{1}{\alpha\pi}.$$

Find c:

$$\frac{f(y)}{g(y)}\le c.$$

Use inverse transform algorithm:

$$G(y)=\int_{-\infty}^{y}A\left(1+\left(\frac{t-\lambda}{\alpha}\right)^2\right)^{-1}dt$$

$$=\int_{-\infty}^{\frac{y-\lambda}{\alpha}}A(1+u^2)^{-1}\alpha du$$

$$=A\alpha\left[\tan^{-1}(u)\right]_{-\infty}^{\frac{y-\lambda}{\alpha}}$$

$$=\frac{1}{\pi}\left(\tan^{-1}\left(\frac{y-\lambda}{\alpha}\right)+\frac{\pi}{2}\right).$$

$$U=\frac{1}{\pi}\left(\tan^{-1}\left(\frac{y-\lambda}{\alpha}\right)+\frac{\pi}{2}\right)$$

$$\pi U=\tan^{-1}\left(\frac{y-\lambda}{\alpha}\right)+\frac{\pi}{2}$$

$$\frac{y-\lambda}{\alpha}=\tan\left(\pi U-\frac{\pi}{2}\right)$$

$$y=\alpha\cdot\tan\left(\pi U-\frac{\pi}{2}\right)+\lambda.$$

$$F^{-1}(U)=\alpha\cdot\tan\left(\pi U-\frac{\pi}{2}\right)+\lambda.$$

$$\frac{K\left(1+\left(\frac{y-\lambda}{\alpha}\right)^2\right)^{-m}e^{-\nu\tan^{-1}\left(\frac{y-\lambda}{\alpha}\right)}}{\frac{1}{\alpha\pi}\left(1+\left(\frac{y-\lambda}{\alpha}\right)^2\right)^{-1}}\le c.$$

$$\frac{d}{dx}\left(-\nu\tan^{-1}\left(\frac{x-\lambda}{\alpha}\right)\right)$$

$$=-\nu\frac{du}{dx}\frac{d}{du}\left(\tan^{-1}u\right)$$

$$=-\nu\cdot\frac{1}{\alpha}\cdot\frac{1}{1+u^2}$$

$$\frac{d}{dy}\left[\alpha\pi K\left(1+\left(\frac{y-\lambda}{\alpha}\right)^2\right)^{1-m}e^{-\nu\tan^{-1}\left(\frac{y-\lambda}{\alpha}\right)}\right]=0.$$

$$\frac{d}{dy}\left[\left(1+\left(\frac{y-\lambda}{\alpha}\right)^2\right)^{1-m}e^{-\nu\tan^{-1}\left(\frac{y-\lambda}{\alpha}\right)}\right]=0.$$

$$(1-m)\left(2\tfrac{1}{\alpha}\left(\tfrac{y-\lambda}{\alpha}\right)^1\right)\left(1+\left(\tfrac{y-\lambda}{\alpha}\right)^2\right)^{-m}\cancel{e^{-\nu\tan^{-1}\left(\frac{y-\lambda}{\alpha}\right)}}$$

$$+\left(1+\left(\tfrac{y-\lambda}{\alpha}\right)^2\right)^{1-m}\left(-\nu\cdot\tfrac{1}{\alpha}\cdot\tfrac{1}{1+\left(\frac{y-\lambda}{\alpha}\right)^2}\right)\cancel{e^{-\nu\tan^{-1}\left(\frac{y-\lambda}{\alpha}\right)}}=0.$$

$$(1-m)\left(2\cancel{\tfrac{1}{\alpha}}\left(\tfrac{y-\lambda}{\alpha}\right)^1\right)\cancel{\left(1+\left(\tfrac{y-\lambda}{\alpha}\right)^2\right)^{-m}}+\left(1+\left(\tfrac{y-\lambda}{\alpha}\right)^2\right)^{1-m}\left(-\nu\cdot\cancel{\tfrac{1}{\alpha}}\cdot\tfrac{1}{1+\left(\frac{y-\lambda}{\alpha}\right)^2}\right)=0.$$

$$(1-m)\left(2\frac{y-\lambda}{\alpha}\right)+\cancel{\left(1+\left(\frac{y-\lambda}{\alpha}\right)^2\right)}\left(-\nu\cdot\cancel{\frac{1}{1+\left(\frac{y-\lambda}{\alpha}\right)^2}}\right)=0,\qquad \left(1+\left(\frac{y-\lambda}{\alpha}\right)^2\right)^{-m}>0$$

$$2(1-m)\frac{y-\lambda}{\alpha}=\nu.$$

$$\frac{y-\lambda}{\alpha}=\frac{\nu}{2(1-m)}$$

$$y=\frac{\nu\alpha}{2(1-m)}+\lambda.$$

Thus,

$$c = \alpha\pi K \left(1+\left(\frac{y-\lambda}{\alpha}\right)^2\right)^{1-m} e^{-\nu\tan^{-1}\left(\frac{y-\lambda}{\alpha}\right)}$$

$$= \alpha\pi K \left(1+\left(\frac{\nu}{2(1-m)}\right)^2\right)^{1-m} e^{-\nu\tan^{-1}\left(\frac{\nu}{2(1-m)}\right)}.$$

Thus, accept if

$$u < \frac{\cancel{K}\left(1+\left(\frac{y-\lambda}{\alpha}\right)^2\right)^{-m} e^{-\nu\tan^{-1}\left(\frac{y-\lambda}{\alpha}\right)}}{\cancel{\alpha\pi K}\left(1+\left(\frac{\nu}{2(1-m)}\right)^2\right)^{1-m} e^{-\nu\tan^{-1}\left(\frac{\nu}{2(1-m)}\right)} \cdot \cancel{\frac{1}{\alpha\pi}}\left(1+\left(\frac{y-\lambda}{\alpha}\right)^2\right)^{-1}}$$

$$= \frac{\left(1+\left(\frac{y-\lambda}{\alpha}\right)^2\right)^{-m} e^{-\nu\tan^{-1}\left(\frac{y-\lambda}{\alpha}\right)}}{\left(1+\left(\frac{\nu}{2(1-m)}\right)^2\right)^{1-m} e^{-\nu\tan^{-1}\left(\frac{\nu}{2(1-m)}\right)}},$$

$+$ $\left(1+\left(\frac{y-\lambda}{\alpha}\right)^2\right)^{-m}$

$$f(y) = K\left(1+\left(\frac{y-\lambda}{\alpha}\right)^2\right)^{-m} e^{-\nu\tan^{-1}\left(\frac{y-\lambda}{\alpha}\right)}$$

$$g(y) = A\left(1+\left(\frac{y-\lambda}{\alpha}\right)^2\right)^{-m}$$

Inverse transform algorithm can't be used. Thus, think of forcing it into a t-distribution.

t-distribution:

$$f(y) \propto \left(1+\frac{y^2}{k}\right)^{-\frac{k+1}{2}}.$$

We have

$$g(y) \propto \left(1+\left(\frac{y-\lambda}{\alpha}\right)^2\right)^{-m}. \quad \text{Let } m=\frac{k+1}{2}, \quad k=2m-1;$$

$$= \left(1+\left(\frac{y-\lambda}{\alpha}\right)^2\right)^{-\frac{k+1}{2}}$$

$$= \left(1+\frac{(y-\lambda)^2 k}{\alpha^2 k}\right)^{-\frac{k+1}{2}}$$

$$= \left(1+\left(\frac{\sqrt{k}(y-\lambda)}{\alpha}\right)^2\frac{1}{k}\right)^{-\frac{k+1}{2}}$$

$$= \left(1+\frac{z^2}{k}\right)^{-\frac{k+1}{2}}, \quad z=\frac{\sqrt{k}(y-\lambda)}{\alpha}$$

$$y = z\cdot\frac{\alpha}{\sqrt{k}}+\lambda.$$

To conclude, use

$$Y = z\cdot\frac{\alpha}{\sqrt{k}}+\lambda.$$

$\downarrow$

generated from t-distribution with degree of freedom $= k = 2m-1$.

$$\frac{f(y)}{g(y)} \leqslant c.$$

$$\frac{K\left(1+\left(\frac{y-\lambda}{\alpha}\right)^2\right)^{-m} e^{-\nu\tan^{-1}\left(\frac{y-\lambda}{\alpha}\right)}}{A\left(1+\left(\frac{y-\lambda}{\alpha}\right)^2\right)^{-m}} \leqslant c.$$

$$\frac{K}{A} e^{-\nu\tan^{-1}\left(\frac{y-\lambda}{\alpha}\right)} \leqslant c.$$

$$\frac{d}{dy}\left[\frac{K}{A} e^{-\nu\tan^{-1}\left(\frac{y-\lambda}{\alpha}\right)}\right] = 0.$$

$$\frac{d}{dy}\left[e^{-\nu\tan^{-1}\left(\frac{y-\lambda}{\alpha}\right)}\right] = 0.$$

$$-\nu \frac{1}{\alpha} \frac{1}{1+\left(\frac{y-\lambda}{\alpha}\right)^2} e^{-\nu\tan^{-1}\left(\frac{y-\lambda}{\alpha}\right)} = 0.$$

$$-\frac{\nu}{\alpha} \frac{1}{1+\left(\frac{y-\lambda}{\alpha}\right)^2} = 0.$$

$$\nu = 0. \quad ?$$

Can't find the maximum using differentiation.

$\frac{K}{A} e^{-\nu\tan^{-1}\left(\frac{y-\lambda}{\alpha}\right)}$ is a monotonous decreasing one, so it's at its maximum when $\nu > 0$: $\tan^{-1}\left(\frac{y-\lambda}{\alpha}\right) = -\frac{\pi}{2}$

$\nu < 0$: $\tan^{-1}\left(\frac{y-\lambda}{\alpha}\right) = \frac{\pi}{2}$,

so that at maximum, $\frac{K}{A} e^{-\nu\tan^{-1}\left(\frac{y-\lambda}{\alpha}\right)} = \frac{K}{A} e^{\frac{\pi}{2}\nu}$.

Hence, we pick $c = \frac{K}{A} e^{\frac{\pi}{2}\nu}$

Accept if

$$U < \frac{\cancel{K\left(1+\left(\frac{y-\lambda}{\alpha}\right)^2\right)^{-m}} e^{-\nu\tan^{-1}\left(\frac{y-\lambda}{\alpha}\right)}}{\cancel{\frac{K}{A}} e^{\frac{\pi}{2}\nu} \cdot \cancel{A\left(1+\left(\frac{y-\lambda}{\alpha}\right)^2\right)^{-m}}}$$

$$= e^{-\nu\tan^{-1}\left(\frac{y-\lambda}{\alpha}\right) - \nu\frac{\pi}{2}}.$$